Written 3

- Due Apr 13 at 10pm
- Points 110
- Questions 5
- Available until Apr 14 at 10pm
- Time Limit None

Instructions

Read chapter 3 of the textbook and complete the following exercises:

- 3.12
- 3.17
- 3.28
- 3.29
- 3.39

You may type your work into the boxes in the quiz or upload individual images or PDFs for each answer. You may not upload a single PDF or PDFs that cover multiple answers. The TAs have complained about the collective disorganization of submissions in past semesters and begged the instructores for this respite (from single PDF uploads).

You must show all work. Answers without work will receive no credit. Answers with insufficient work will receive little to no credit

Please keep your work neat and organized. The TAs will penalize your grade if they are forced to expend undue effort to decipher cryptic submissions.

Quiz Type

This quiz was locked Apr 14 at 10pm.

Attempt History

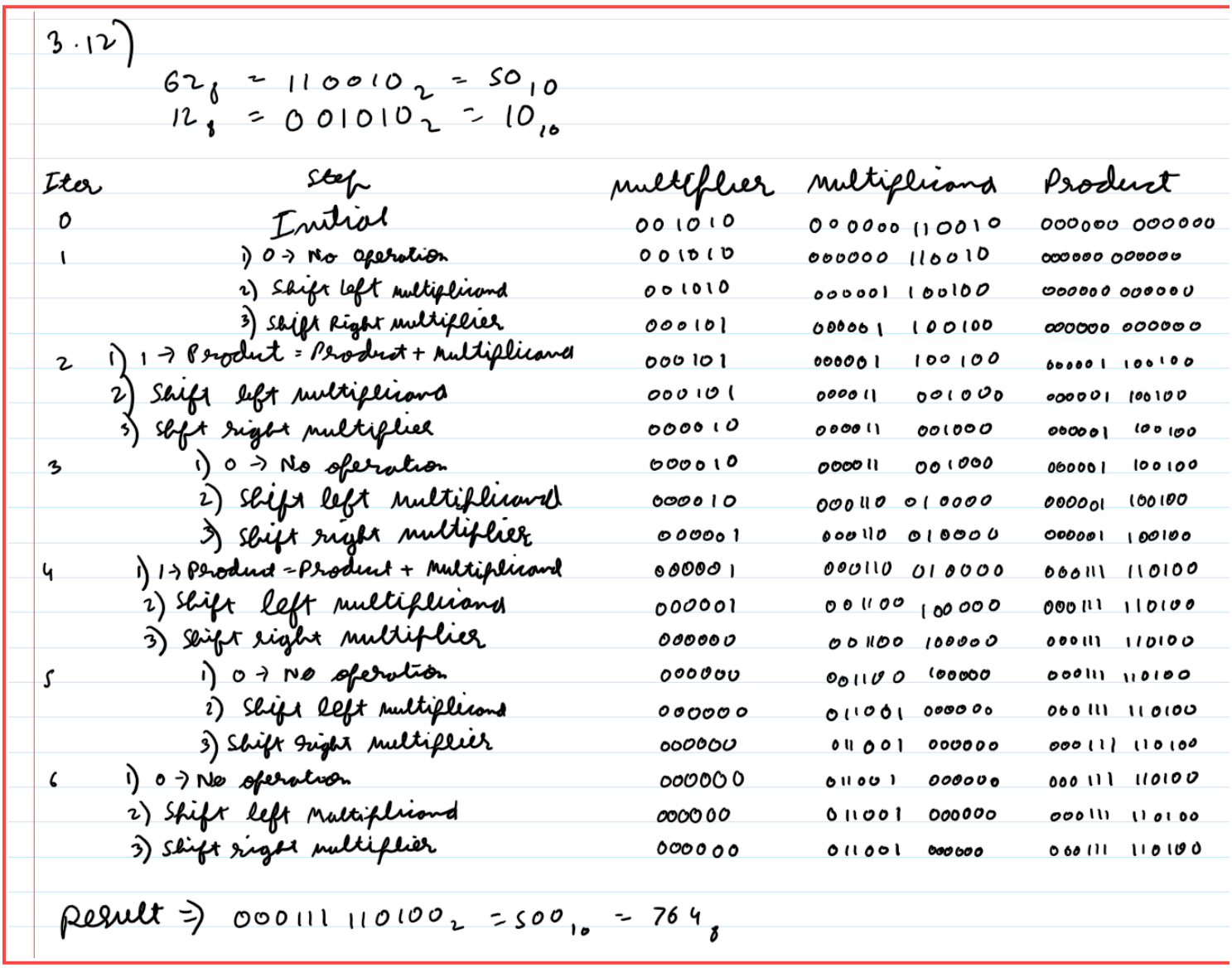
	Attempt	Time	Score
LATEST	Attempt 1	13 minutes	100 out of 110

Score for this quiz: 100 out of 110
Submitted Apr 13 at 9:49pm
This attempt took 13 minutes.

Question 1
18 / 20 pts

3.12

Your Answer:



3.12)

$62_8 = 110010_2 = 50_{10}$

$12_8 = 001010_2 = 10_{10}$

Iter	Step	multiplier	multiplicand	Product
0	Initial	001010	000000 110010	000000 000000
1	1) 0 → No operation	001010	000000 110010	000000 000000
	2) Shift left multiplicand	001010	000001 100100	000000 000000
	3) Shift Right multiplier	000101	000001 100100	000000 000000
2	1) 1 → Product = Product + multiplicand	000101	000001 100100	000001 100100
	2) Shift left multiplicand	000101	000011 001000	000001 100100
	3) Shift right multiplier	000010	000011 001000	000001 100100
3	1) 0 → No operation	000010	000011 001000	000001 100100
	2) Shift left multiplicand	000010	000110 010000	000001 100100
	3) Shift right multiplier	000001	000110 010000	000001 100100
4	1) 1 → Product = Product + multiplicand	000001	000110 010000	000111 110100
	2) Shift left multiplicand	000001	001100 100000	000111 110100
	3) Shift right multiplier	000000	001100 100000	000111 110100
5	1) 0 → No operation	000000	001100 100000	000111 110100
	2) Shift left multiplicand	000000	011001 000000	000111 110100
	3) Shift right multiplier	000000	011001 000000	000111 110100
6	1) 0 → No operation	000000	011001 000000	000111 110100
	2) Shift left multiplicand	000000	011001 000000	000111 110100
	3) Shift right multiplier	000000	011001 000000	000111 110100

Result ⇒ $000111\ 110100_2 = 500_{10} = 764_8$

final multiplicand register value is incorrect -2

Question 2

20 / 20 pts

3.17

Your Answer:

3.17) $0x33 = 51_{10}$; $0x55 = 85_{10}$

$51 \rightarrow 3 \times 17 = (2+1)(2 \times 2 \times 2 \times 2 + 1)$

1) $x = 0x55 + (0x55 << 1) = 85 + 170 = 255$

2) Result $= x + (x << 4) = 255 + 4080 = 4335$

Ans) $4335_{10} = 0x10EF$

2 shift and 2 addition. Nice!

Question 3

20 / 20 pts

3.28

Your Answer:

3.28) Format : 16 bit fraction, 8 bit frac extension = 24 bit two's compliment frac then 7 bit magnitude, 1 bit sign exponent. No hidden 1.

1) Normalizing $\rightarrow -0.15625 = -0.625 \times 2^{-2}$

2) +ve $= 0.625 = 0.101000\ 00000000\ 00000000$

Compliment $= 1.011000\ 00000000\ 00000000$

Bits 1-16 : 1011 0000 0000 0000

Bits 17-24 : 0000 0000

3) Magnitude $= 2 = 0000010$; Sign $= 1 \rightarrow 0000\ 0101$

∴ Ans) 1011 0000 0000 0000 0000 0000 0000 0101

Comparison to IEEE 754 single :-

1) Has a similar precision to IEEE 754 single [24 bit two's compliment vs 23 bits]

2) 7 bit magnitude exponent is similar range to IEEE 754 single.

Disadvantage :- No NaN, No infinity, No denormals, Weird sign-magnitude expo (sign on right)

Question 4

17 / 20 pts

3.29

Your Answer:

3.29) $A = 26.125 = 1.1010001000 \times 2^4 \rightarrow 0\ 10100\ 1010001000$

$B = 4.150390625 \times 10^{-1} = 1.0101001001 \times 2^{-2} \rightarrow 0\ 01110\ 1010100100$

Aligning → diff = 6, shift B right 6 : B → 0.0000011010 ; G = 1 ; R = 0 ; S = 1

Adding → $1.1010001000 + 0.0000011010 = 1.1010100010$

Rounding → G = 1 ; S = 1 → Round Up → 1.1010100011

Ans) 0 10100 1010100011 = 26.546875_{10}

Exponent bias of 16 is used instead of 15, producing biased exponents of 20 = 10100 for A and 14 = 01110 for B instead of the correct 19 = 10011 and 13 = 01101. Final encoding is incorrect (-3)

Question 5

25 / 30 pts

3.39

Your Answer:

3.39) $1.666015625 = 1.1010101010 \times 2^0 \rightarrow 0\ 10000\ 1010101010$

$19760 = 1.0011010011 \times 2^{14} \rightarrow 0\ 11110\ 0011010011$

$-19744 = -1.0011010010 \times 2^{14} \rightarrow 1\ 11110\ 0011010010$

$P1 = 1.666015625 \times 19760 = 32920.46875 \rightarrow 32928 \rightarrow 0\ 11111\ 0000000101$

$P2 = 1.666015625 \times (-19744) = -32893.8125 \rightarrow -32896 \rightarrow 1\ 11111\ 0000000100$

$P1 + P2 \rightarrow 1.0000000101 - 1.0000000100 = 0.0000000001 \times 2^{15} \rightarrow$ Normalizing $\rightarrow 1.0 \times 2^5$

Ans) 0 10101 0000000000 = 32_{10}

But the actual value is $1.666015625 \times 16 = 26.65625_{10}$

→ Catastrophic cancellation :- Rounding errors in P1 and P2 are larger than the true result, so all significant bits are lost when the two products are subtracted.

Too direct approach, question explicitly asks to show all steps in 16 bit binary format. Conversion, multiplication and rounding steps are missing.

Quiz Score: 100 out of 110